

FIT3077 Software Engineering: Architecture and Design

Team name: Santorini Craftsmen 001

Team leader: Emil Chin Jia Zhen (33520542)

Team members: 1. Tiffany Lau Kho Jen (33521859)

2. Pee Yee Peen (33591032)

3. Wong Xin Thung (33592357)

1.0 Team Information and Technology Prototypes.....	3
1.1 Team Name and Team Photo.....	3
1.2 Team Membership.....	4
1.2.1 Basic Personal Information.....	4
1.2.2 Team Leader Selection.....	4
1.2.3 Technical and Professional Strengths.....	4
1.2.4 Fun Fact.....	5
1.3 Team Schedule.....	6
1.4 Technology Stack and Justification.....	7
1.4.1 Selection of Programming Language.....	7
1.4.2 Technical Prototyping.....	7
2.0 User Stories.....	7
3.0 Domain Model.....	7

1.0 Team Information and Technology Prototypes

This section shows the team information, including the team name, team photo, team membership, and team schedule. Technology stack and justification are also included in this section.

1.1 Team Name and Team Photo

Team Name: Santorini Craftsmen 001

Moodle Team Name: MA_Tuesday06pm_Team001

Team Photo:



Figure 1.1.1 Santorini Craftsmen's Team Photo

The team members of Santorini Craftsmen are arranged as follows:

Leftmost: Wong Xin Thung; Middle-left: Pee Yee Peen; Middle-right: Tiffany Lau Kho Jen;
Rightmost: Emil Chin Jia Zhen

1.2 Team Membership

This section provides an overview of each team member's basic personal information, such as name and contact details. It also includes the selection of a team leader, along with a list of each team member's technical and professional strengths, as well as a fun fact description.

1.2.1 Basic Personal Information

Table 1.2.1 shows the personal details of each team member, including name, student ID, student email, and contact number.

Table 1.2.1 Basic information about each team member

Team Members Name	Contact details		
	Student ID	Student Email	Contact Number
Tiffany Lau Kho Jen	33521859	tlau0015@student.monash.edu	+60107704147
Pee Yee Peen	33591032	yee0001@student.monash.edu	+601112733288
Wong Xin Thung	33592357	xwon0021@student.monash.edu	+601155190067
Emil Chin Jia Zhen	33520542	echi0039@student.monash.edu	+60143988417

1.2.2 Team Leader Selection

The team leader chosen for Sprint 1 is **Emil Chin Jia Zhen (33520542)**.

1.2.3 Technical and Professional Strengths

This section introduces the technical and professional aspects of each team member.

Member 1: Tiffany Lau Kho Jen

I am proficient in coding several programming languages such as Python, Java, HTML, CSS, and JavaScript. My expertise in Python and Java allows me to build interactive games which include implementing the game logic and applying the object-oriented programming concepts. I also utilised the frameworks and libraries available to create these games like Pygame to render graphics such as sprites, sprites animations, and the background. Moreover, I am skillful in developing responsive web-based applications, using a combination of HTML, CSS, and JavaScript. I used HTML and CSS to create user-friendly interfaces that are visually appealing and easy to navigate through. I leverage Javascript mostly to handle user interactions with event listeners as well as real-time content updates. I enjoy tackling difficult problems with an analytical mindset, allowing me to expand my knowledge and technical expertise in this field.

Member 2: Pee Yee Peen

I have extensive experience writing code across multiple languages, including Python, Java, MATLAB, SQL, and HTML. I am proficient in developing efficient applications. For instance, I utilise Python for scripting and Java for software development. Besides, I have a strong foundation in solving numerical problems and ensuring effective database management by using MATLAB and SQL respectively. I am also skilled in structuring web content using HTML. This diverse skill set allows me to adapt to different programming challenges and efficiently develop solutions accordingly.

Member 3: Wong Xin Thung

I have previously studied Python, Java, and MATLAB, among other programming languages, and have used them to finish some of my assignments. For example, I previously developed software using Python, and for my last semester assignment, I utilised Java to create a text-based game. At the same time, I am familiar with GitLab, which may enhance team collaboration efficiency by efficiently sharing and managing code with peers. Along with knowing the Agile methodology, I believe this approach can put it into practice while doing my assignments, which helped me better grasp how to move the project along and adjust to the pace of cooperation.

Member 4: Emil Chin Jia Zhen

I have a basic understanding of programming, mainly in Python and Java. Besides, I used Java to create a simple game and Python to code a game simulation, along with other assignments in previous semesters. I also know some backend coding, like working with databases and JavaScript, though I am still improving in these areas. Additionally, I have used MATLAB to solve physics and math problems and MARIE to create a simple countdown display. My strengths include problem-solving, logical thinking, and adaptability, and I am keen to keep learning and improving my skills in software and game development.

1.2.4 Fun Fact

This section highlights a unique and lesser-known fun fact about each team member.

Member 1: Tiffany Lau Kho Jen

A fun fact about me is that I like to listen to music composed of rap, hip-hop and R&B genres. Listening to them lifts my mood up instantly and helps me to stay relaxed. Besides that, it improves my focus when studying or exercising as I become more energetic and active. I also attended one of my favourite rapper's concerts two years ago in Kuala Lumpur so that I could see him perform live on stage and gain an unforgettable experience.

Member 2: Pee Yee Peen

A fun fact about me is that I am a spicy enthusiast. I enjoy exploring different cuisines that bring the heat, as I find that spicy food not only excites my taste buds but also helps me relax when I am feeling stressed. I enjoy every moment of the spice challenge.

Member 3: Wong Xin Thung

One interesting thing about me is that I enjoy watching action, suspense, and Disney films. I find that watching films helps me relax and reduce my stress, especially comedies that always make me laugh out loud and help me forget about my problems. Every movie is like a fantastic adventure that lets me lose myself in it and enjoy each moment, whether it has a thrilling story or a heartwarming and inspiring story.

Member 4: Emil Chin Jia Zhen

A fun fact about me is that I always do better on the second try. For example, when I try to throw rubbish into a bin from a distance, I almost always miss it on the first attempt but get it in on the second. This is a fact—it has been proven time and time again that I tend to perform better on my second try. So, bear with me! XD

1.3 Team Schedule

Our team has set up a regular meeting schedule to keep communication flowing and ensure we stay on the same page. We'll be meeting **twice a week**, on **Tuesdays** and **Wednesdays** at **10:00 AM**, for about **an hour**. Depending on everyone's availability, these meetings will be held either in **person or online**, so all team members can join no matter where they are. If we are approaching a deadline or need to discuss something urgent, we will schedule **extra meetings** if needed. Each session will have a clear agenda covering progress updates, task assignments, and any roadblocks we're facing. We'll also take notes and send it to our WhatsApp group, and make meetings minutes after every meeting to keep track of action items and make sure nothing slips through the cracks.

When it comes to our regular work schedule, everyone is expected to contribute at least **10 hours per week** to the project. We will distribute tasks based on each person's strengths, availability, and expertise to make sure the workload is balanced. To stay organised, we will be using **Trello** to track tasks, deadlines, and overall progress, **Gitlab** to commit tasks done by each member. Assignments will be discussed and updated during our meetings, and everyone will be responsible for keeping their status current. If anyone runs into challenges, they're encouraged to ask for help so we can tackle issues together and keep things moving forward. By staying in sync and maintaining a structured approach, we aim to complete our work efficiently and on time.

1.4 Technology Stack and Justification

1.4.1 Selection of Programming Language

We made this decision as a team after carefully evaluating the project specifications and our team's technical strengths.

1.4.1.1 Java and Java Swing

Our team has decided to select Java as our chosen programming language together with its graphical user interface (GUI) library called Java Swing, especially for the development of the Santorini board game.

One of the main reasons for choosing Java is that all team members have experience coding in the language itself throughout our past assignments and personal projects. We are familiar with object-oriented programming (OOP) in Java which consists of several concepts such as polymorphism, inheritance, encapsulation, and abstraction. Besides that, we learned how to write efficient and maintainable code through the application of the SOLID principles, hence contributing to the enhancement of the software design and its robust system. We believe that all these concepts, skills, and other relevant knowledge are essential for the Santorini game development, considering the fact that the game is highly strategic with its comprehensive rules and complex mechanics. The game could also possibly have several extensions such as unique god powers, customizable board features, and the accommodation of multiple players. As such, the aforementioned concepts and principles enable us to make our game design modular and flexible without affecting the overall game logic and its core functionalities. We can also use suitable or appropriate design patterns in Java to organise our code neatly by dividing it into several components with focused responsibilities.

In addition, after doing some research and having group discussions, we agreed that we could fully utilise Java Swing for graphics rendering and animation in our project given its extensive set of components and layout managers (Prepbytes, 2023). The lightweight components including buttons, menus, and labels can be customised and they are easy to use (Prepbytes, 2023). Meanwhile, a variety of layout managers are provided to allow flexible arrangement of components on a GUI and build diverse, visually appealing user interfaces (Prepbytes, 2023). Not only that, Java Swing produces a more intuitive GUI through its robust event handling model to handle user interactions and inputs such as mouse clicks and keyboard presses to control the game elements, menus, and characters (Prepbytes, 2023). Furthermore, the Swing toolkit consists of a "pluggable look and feel" feature that allows us to customise the appearance and behaviour of the components in the GUI (Prepbytes, 2023). This provides a consistent user interface that is easy to navigate across numerous platforms.

As all team members are familiar with the Java programming language and its OOP concepts, it is not very likely that any one of us would require support from the teaching team. Regarding learning how to use the GUI, Java Swing for our project, we will do the necessary research and watch online tutorials to help us develop the essential components

needed for an interactive, scalable Santorini game application with an attractive user interface.

1.4.1.2 Python and Tkinter

We also considered another alternative which is Python along with its GUI called Tkinter. The main reason we chose Java over Python is because we have less experience coding with Python especially when it comes to game development. Although Python does support OOP, we have limited exposure to designing complex class hierarchies and applying object-oriented principles in our program. In comparison to Java, we are not very familiar with the OOP practices in Python such as abstract classes and method overriding as well as its syntax. This could cause us to work less efficiently and slow down our progress when coding the game for our project. Besides that, it may be harder for us to debug our code if multiple errors were encountered due to our lack of knowledge and experience in understanding the game logic and OOP patterns implemented in Python. Regarding its GUI, Tkinter, it has a few limitations as compared to Java Swing. Its outdated default widgets could make the GUI look less visually appealing and modern (Takle, 2025). Styling could appear to be inconsistent across the platforms (Takle, 2025). Tkinter offers limited support in customising the widgets or designing complex layouts due to its lack of advanced layout managers, which causes flickering and less responsiveness of user interfaces (Takle, 2025). Hence, learning how to overcome these limitations could consume time instead of focusing on implementing the essential game logic and design.

1.4.1.3 Final Selection

In conclusion, the team's final choice of programming language that will be used is **Java** and its GUI framework, namely **Java Swing**. All team members have prior exposure to Java and possess a solid understanding of the OOP concepts which are crucial to design a modular, maintainable architecture, given the game's complexity and extensibility. Java Swing also provides more flexibility and functionality than Tkinter due to its large set of components, advanced layout managers, and a reliable event-handling model which supports the development of a dynamic and engaging user interface. We considered using Python with Tkinter, however, the limited experience in Python's OOP and Tkinter's limited features for GUI development would delay the team's productivity in creating the game. Thus, Java is the most suitable choice for this project, as it aligns with our team's skill set and supports maintainable architecture, allowing us to build a stable, high-quality game while prioritizing user experience.

1.4.2 Technical Prototyping

2.0 User Stories

3.0 Domain Model

4.0 Basic UI Design